

항공통신·무선험법 시스템

Communication & Radio Navigation System in Aviation

정비훈련

강사 : 이 연달

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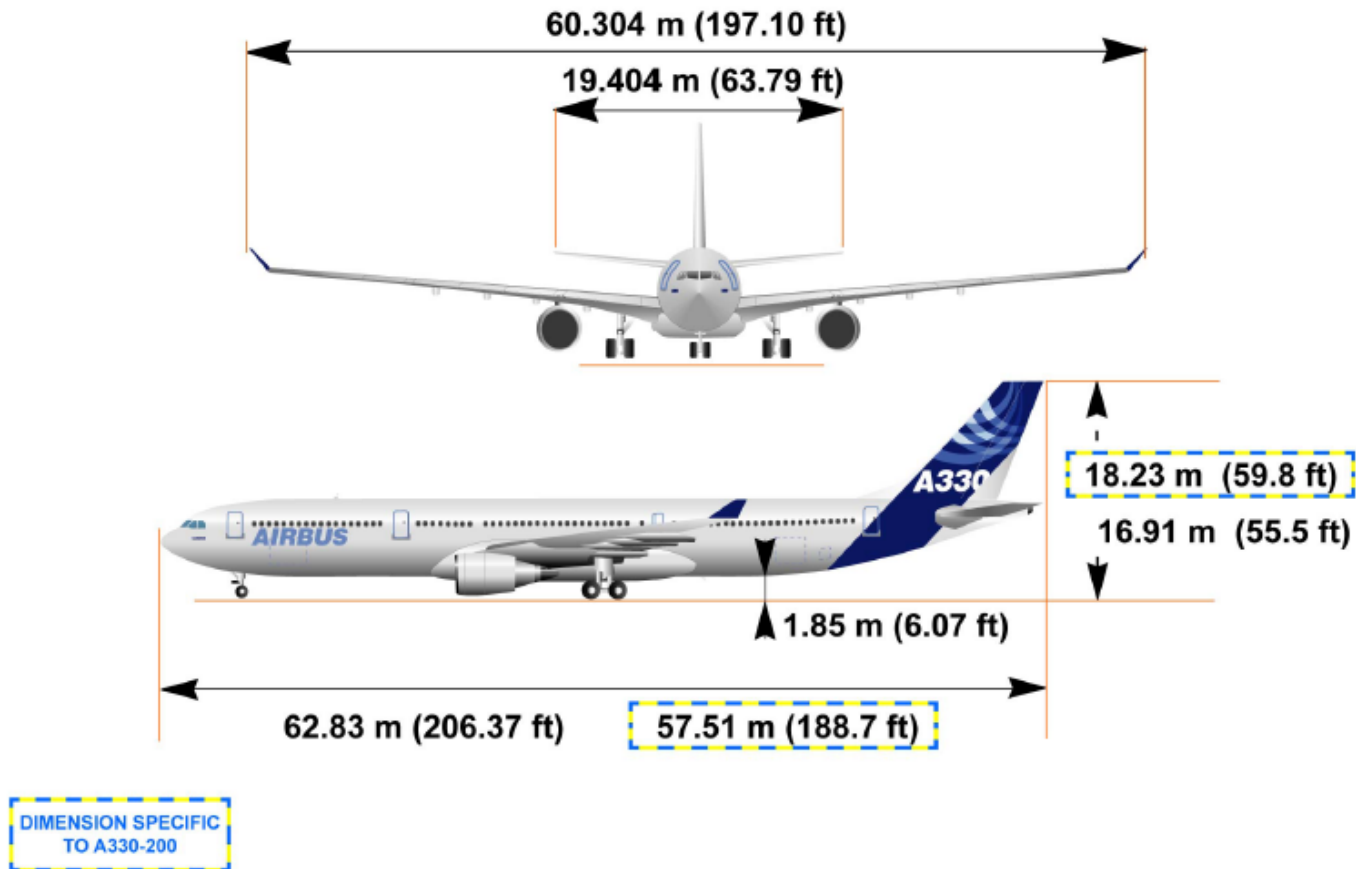
❖ A330 항공기 - 강의 기준 기종



종류	광동체형 제트 여객기
첫 비행	1992년 11월 2일
도입 시기	1994년 1월
현황	생산 중
주요 사용자	• 대한항공 • 에어아시아 X • 중국남방항공 • 중국동방항공 • 터키 항공 • 아시아나항공 • 캐세이퍼시픽 항공
생산 시기	1992년 ~
생산 대수	2019년 3월 31일 기준 1,452대 ^[1]
단가	A330-200: US\$238.5M (2018) A330-300: US\$264.2M (2018) A330-200F: US\$241.7M (2018)
파생형	에어버스 A330 MRTT, 에어버스 A330neo



❖ A330 항공기



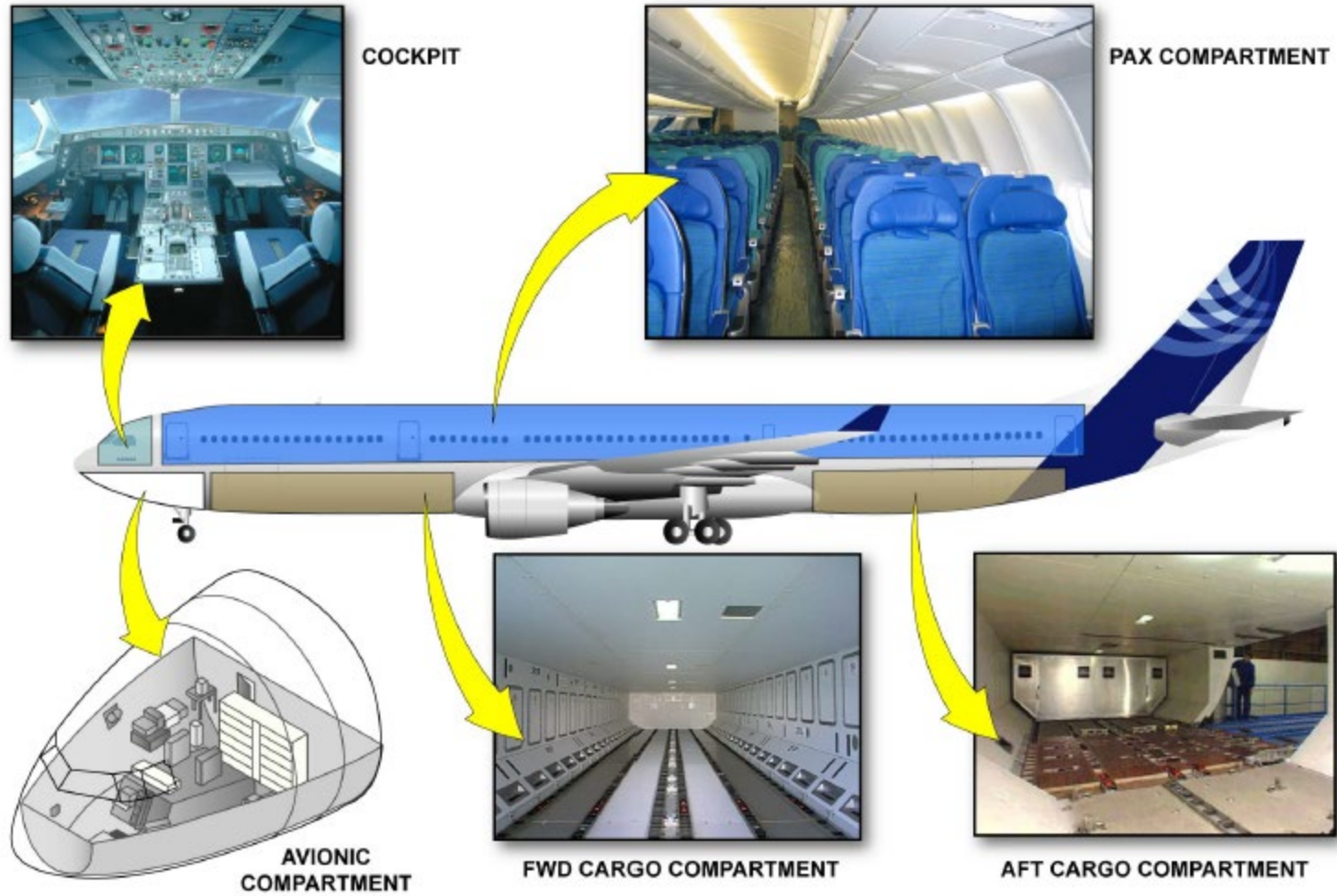
A/C DIMENSIONS

항공기 치수

COCKPIT PANELS PRESENTATION



❖ A330 항공기



GENERAL – MAIN COMPARTMENTS

❖ 23 COMMUNICATION SYSTEM (A330)

- The communications system is composed of:

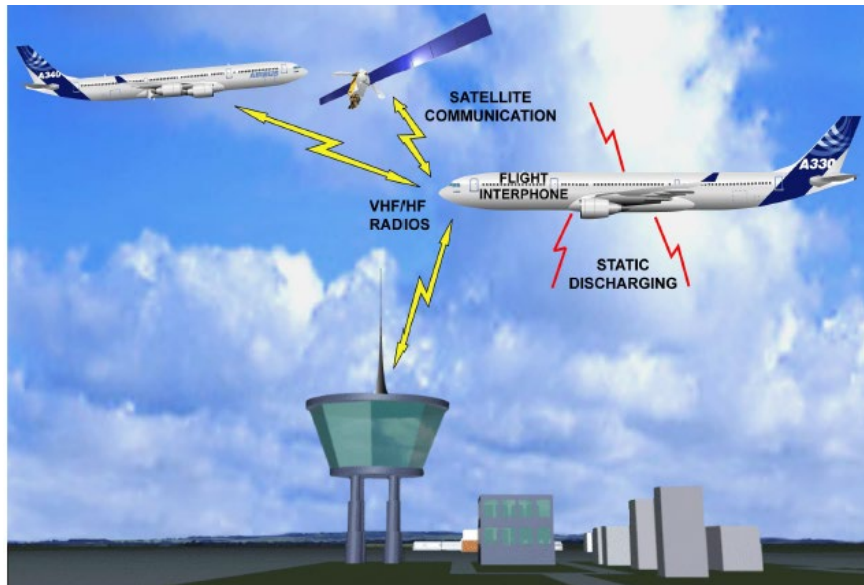
- **Cockpit communications system**,
- cabin communications system,
- **SELCAL system**
- **Cockpit Voice Recorder (CVR) system.**

A. COCKPIT COMMUNICATIONS

There are:

- **two High Frequency (HF) systems** for long range communication,
- **three Very High Frequency (VHF) systems** for short-range line of sight communication,
- a satellite communication system.
- **SElective CALing (SELCAL) system**

B. Emergency Equipment System



Digital Flight Data Recorder (DFDR)

❖ 23 COMMUNICATION SYSTEM (A330)

A. COCKPIT COMMUNICATIONS

1) Very High Frequency (VHF) systems

The VHF system allows short distance voice communications between different A/C (in flight or on ground), or between the A/C and a ground station. The basic version is equipped with 3 VHF Data Radio (VDR) transceivers used for voice communications.

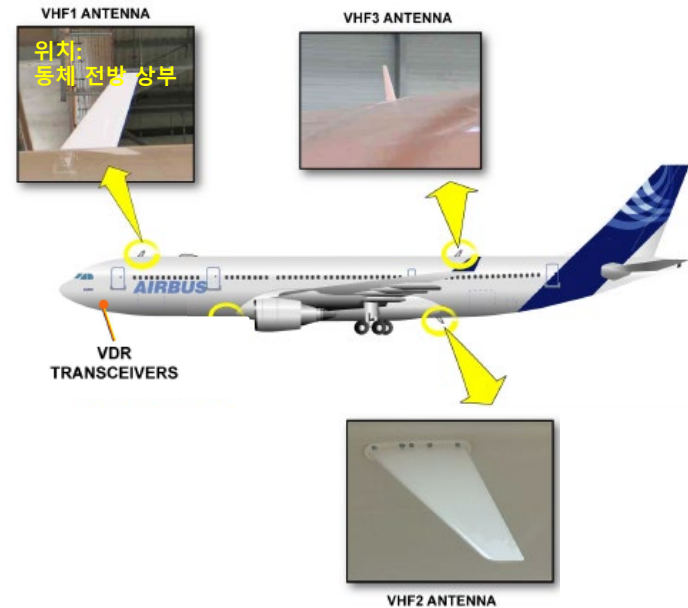
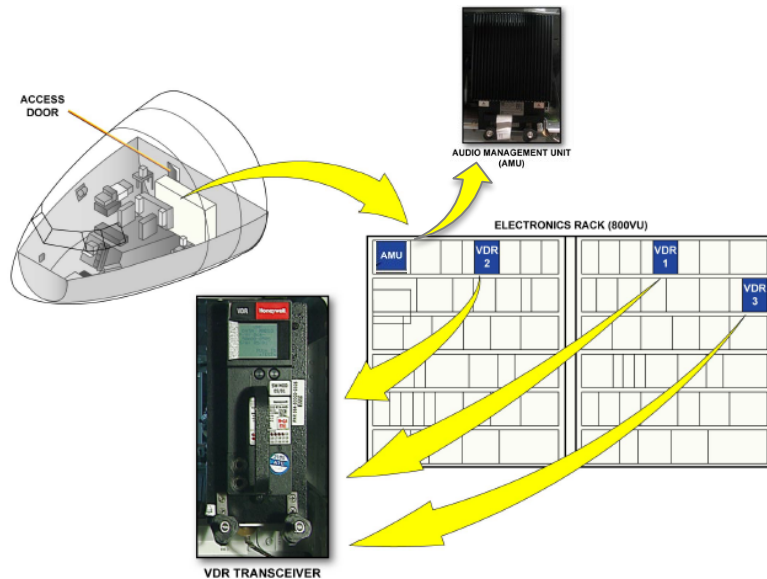
The VDR 3 transceiver is also connected to the Air Traffic Service Unit (ATSU) for direct exchange of data between the aircraft and a ground network, for example aircraft reports or weather reports.

The VHF system comprises:

- 3 VDR transceivers,
- 3 blade antennae,
- the Audio Management Unit (AMU).

The SElective CALLing (SELCAL) system gives visual and aural indications to the flight crew, concerning calls received through the HF system.

NOTE: The SELCAL function is integrated in the AMU.



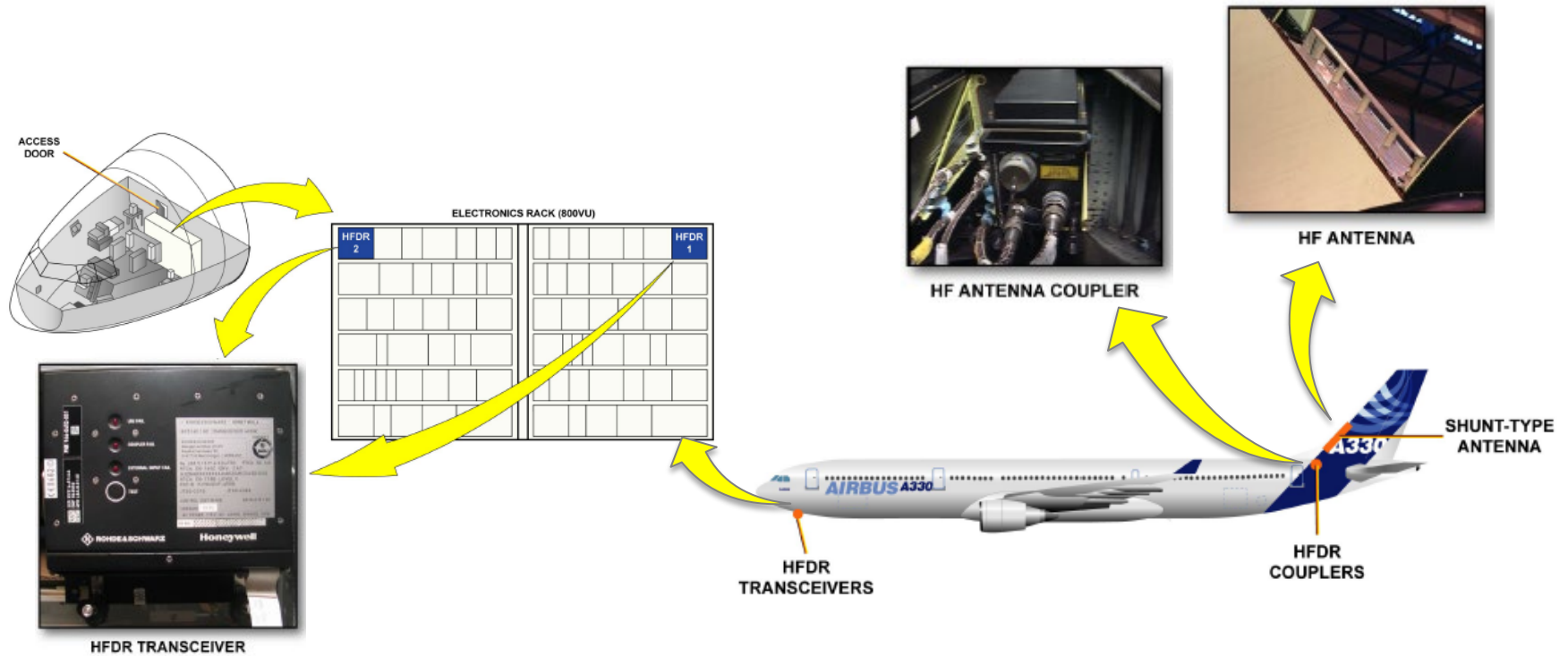
❖ 23 COMMUNICATION SYSTEM (A330)

A. COCKPIT COMMUNICATIONS

2) High Frequency (HF) systems

The High Frequency (HF) system allows long distance voice communications between A/C (in flight or on ground), or between the A/C and a ground station. The HF system is comprised of:

- 2 High Frequency Data Radio (HFDR) transceivers,
- 2 HFDR couplers,
- 1 shunt-type antenna.

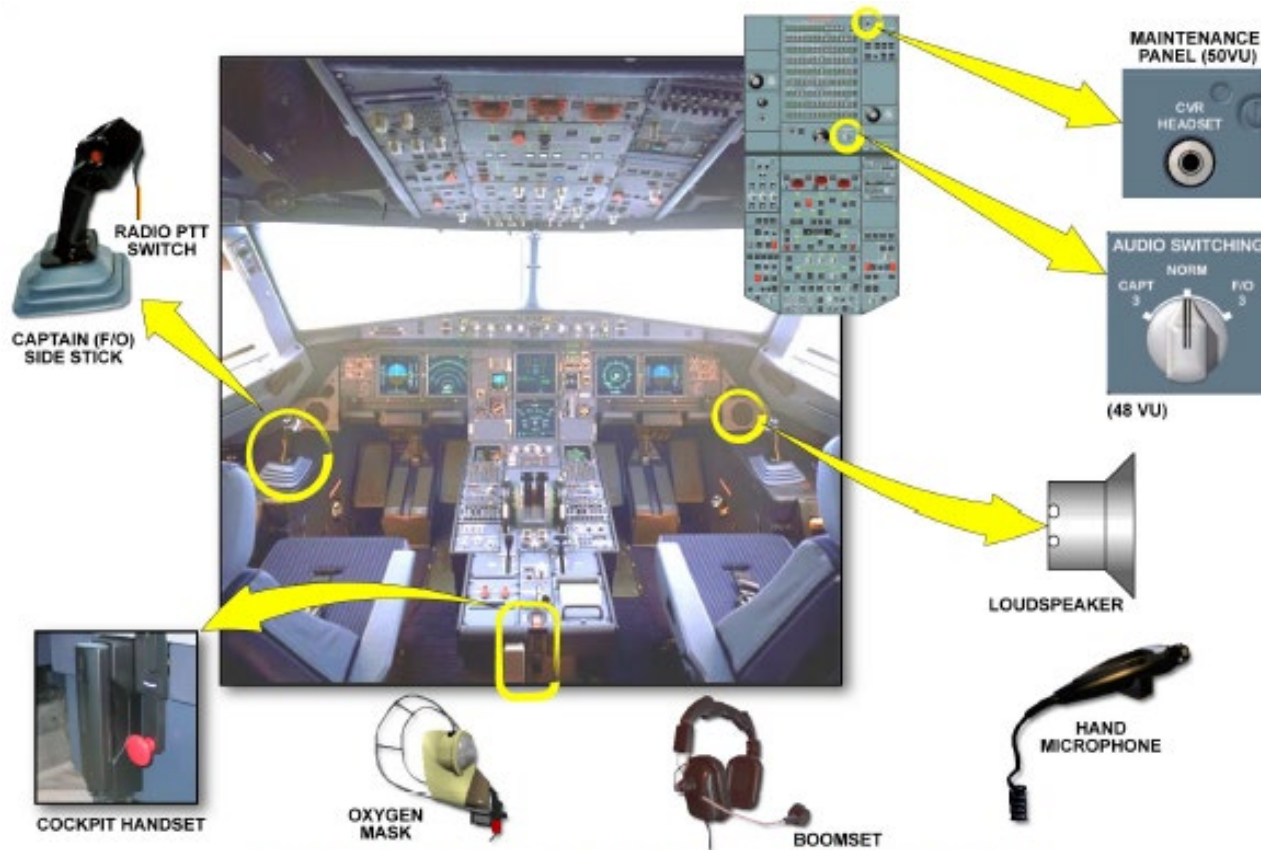


❖ 23 COMMUNICATION SYSTEM (A330)

A. COCKPIT COMMUNICATIONS

3) Cockpit Peripherals & Acoustic Equipment

- The acoustic equipment, which is composed of 2 side-stick radio push to talk, 2 loudspeakers, 3 oxygen mask microphones, facilities for boomsets, headsets and microphones..



COMPONENT LOCATION - COCKPIT PERIPHERALS & ACOUSTIC EQUIPMENT

❖ 23 COMMUNICATION SYSTEM (A330)

A. COCKPIT COMMUNICATIONS

4) Control Panel Operation

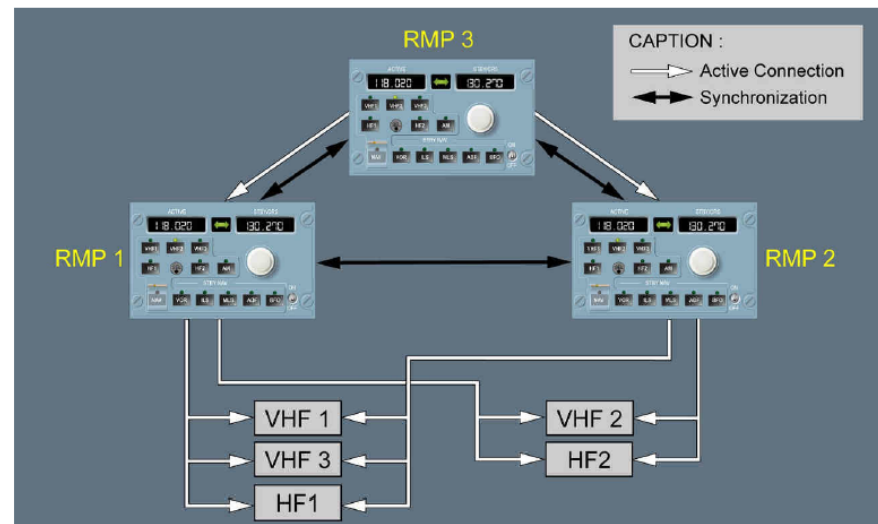
- the Radio Management Panels (RMPs),
The RMPs give the capability of tuning all communication radios. The frequencies are tuned by using the 2 concentric knobs. There are two RMPs installed on the center pedestal and a third one on the overhead panel. Any RMP can tune any communication radio.
- the Audio Control Panels (ACPs),
The ACPs control the channel selection and the volume of transmission/reception. There are three ACPs installed next to an RMP.



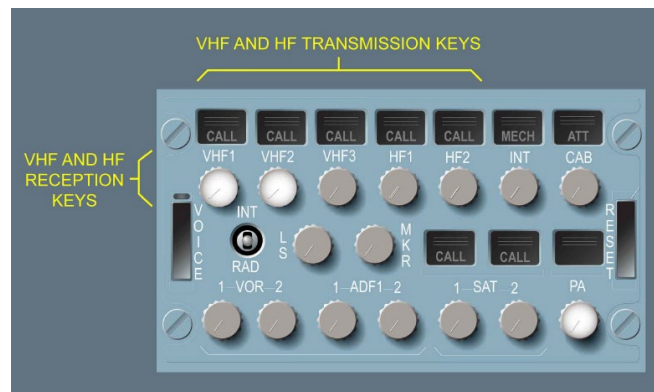
❖ 23 COMMUNICATION SYSTEM (A330)



- **FREQUENCY RANGE:**
- 118.000Mhz to 137.975Mhz,
with 8.33Khz spacing between channels.
- **FREQUENCY RANGE:**
- 2.0000 to 29.9999 MHz



If two RMPs fail, the remaining one controls all the VHF and HF transceivers.



The associated communication means (i.e. VHF 1(2)(3), HF 1(2)) is in transmission mode. When pressed, the transmission key turns off the transmission mode of the associated channel.



The associated communication means (i.e. VHF 1(2)(3), HF 1(2)) is not in transmission mode. When pressed, the transmission key turns on to transmission mode of the associated channel.



Flashes to indicate an incoming of SELCAL. When a SELCAL is received, a buzzer sounds in the cockpit until it is reset, via SELCAL/Call RESET Key.

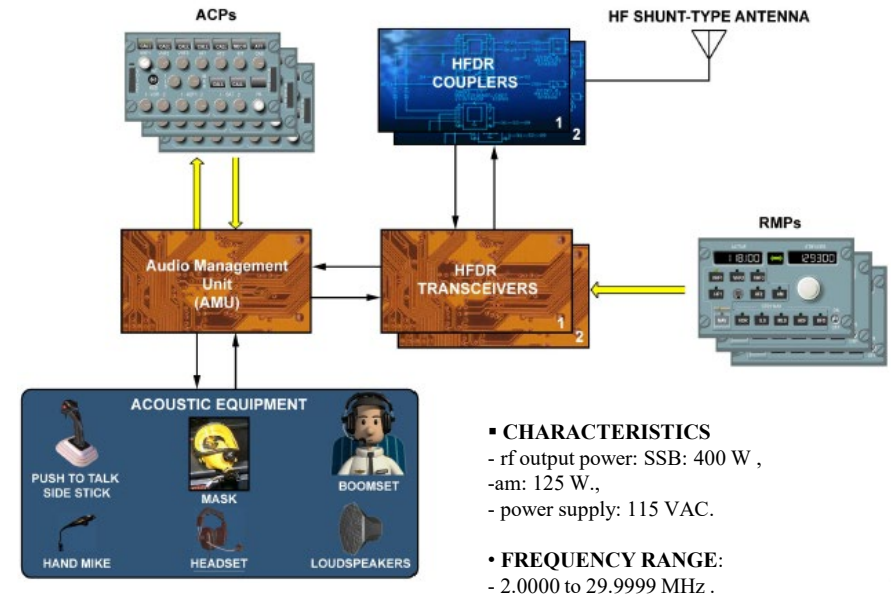
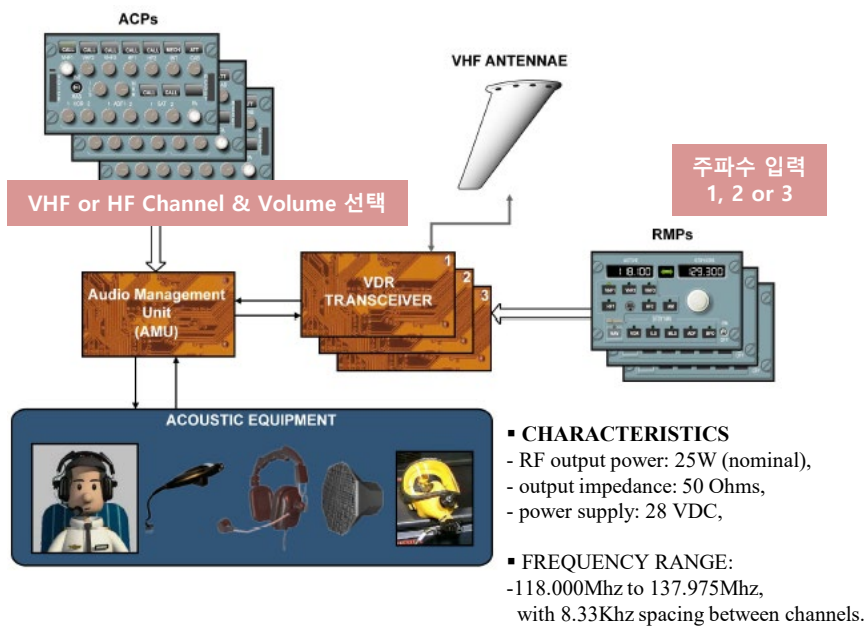
❖ 23 COMMUNICATION SYSTEM (A330)

A. COCKPIT COMMUNICATIONS

5) Systems Operation

The operation of the VHF & HF systems :

- the RMPs are used for VDR and HF frequency control.
- The AMU acts as an interface between the users and the VHF & HF systems for transmission and reception of audio signals.
- The Push -to-talk (PTT) key line is a ground signal sent to the transceivers through the AMU.
- The ACPs are used for VHF & HF transmission or reception selection mode and control of the received audio signal levels through the AMU.
- The SElective CALling (SELCAL) system gives visual and aural indications to the flight crew, concerning calls received through the HF system.



❖ 23 COMMUNICATION SYSTEM (A330)

B. Emergency Equipment System

1. Solid State Cockpit Voice Recorder(SSCVR)_ 조종실 음성 기록장치

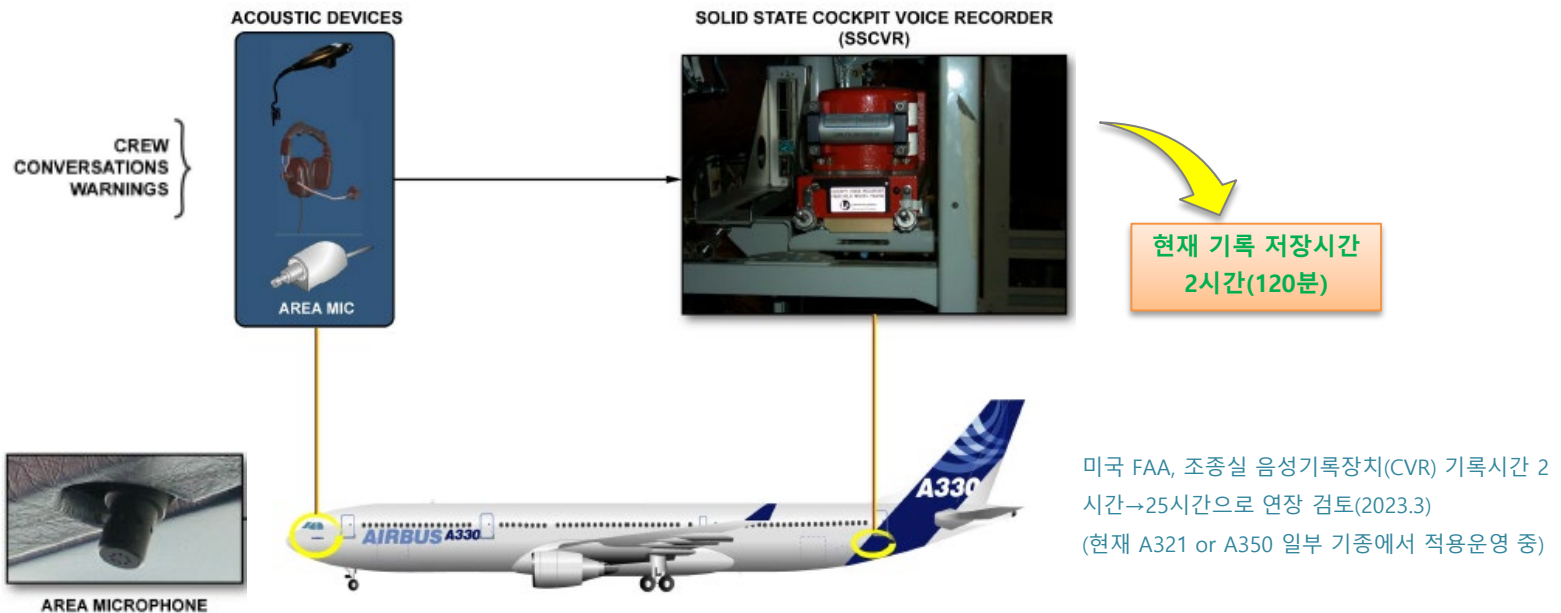
1) Component Location

The Solid State Cockpit Voice Recorder (SSCVR) is in the aft pressurized zone, in the bulk compartment.
The area microphone is in the middle of the windshield, immediately below the overhead panel

2) Principle

The Solid State Cockpit Voice Recorder (SSCVR) is designed to record crew conversations and communications on SOLID STATE MEMORIES, in flight or on ground, and to preserve them in case of an A/C accident.

The recordings are made in SOLID STATE MEMORIES which basically provide a 30-minute recording of continuous operation, or **2 hours with memory extension (+ 90 minutes)**. All data recorded 30 minutes or **2 hours** before is automatically erased by the recording of new data. (2016년 1월1일부터 적용)



SSCVR COMPONENT LOCATION

❖ 23 COMMUNICATION SYSTEM (A330)

B. Emergency Equipment System

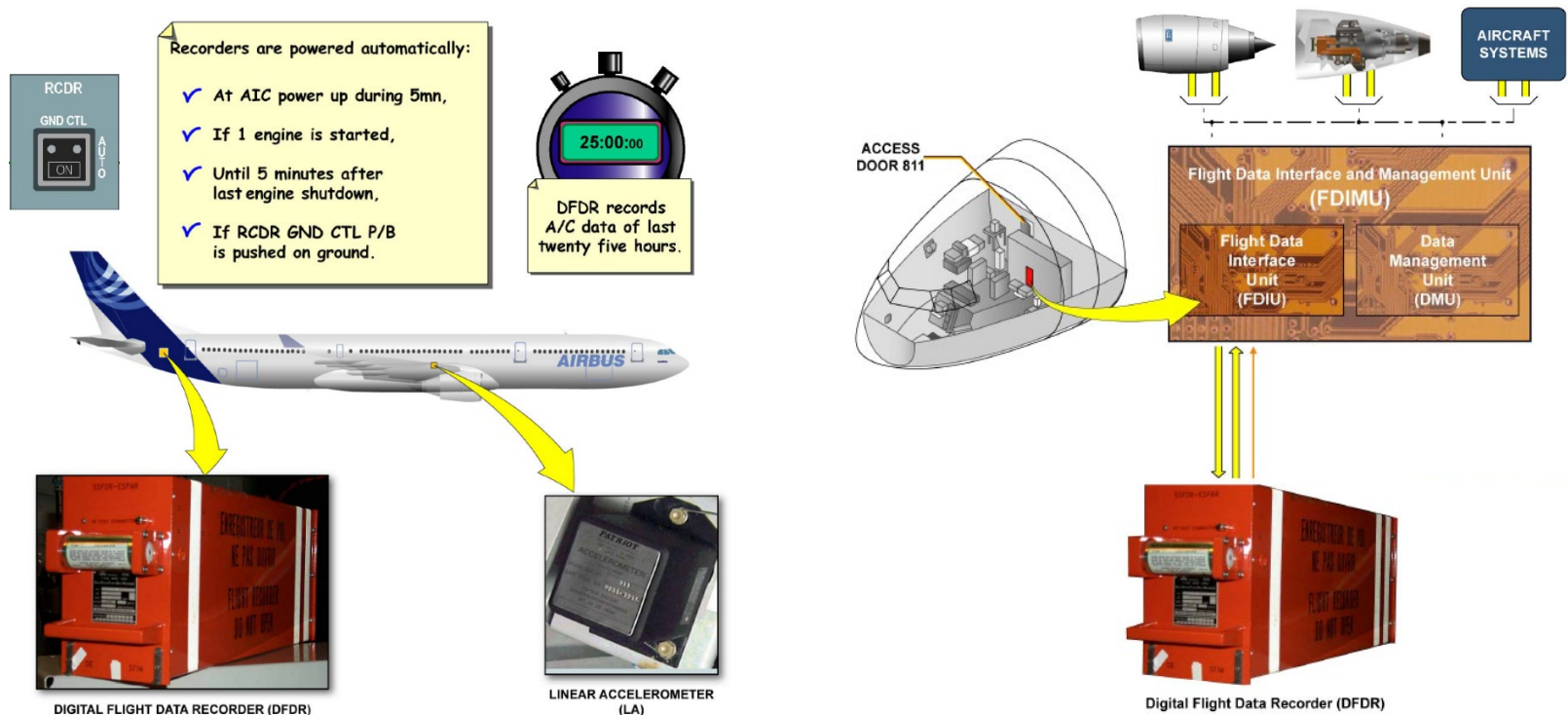
2. Digital Flight Data Recorder(DFDR)_ 디지털 비행 데이터 기록장치

1) Component Location

The DFDR is a recorder installed in the aft part of the aircraft, access through the THS door. The memory board is installed in a protected box. The LA, which measures the acceleration of the aircraft in all three axes, is installed near the gravity center of the aircraft.

2) Principle

The FDIU part of the FDIU, located in the avionics bay, is a computer which collects various basic aircraft system parameters and converts them by internal processing. The aircraft system parameters are transmitted to the DFDR for recording.



❖ 23 COMMUNICATION SYSTEM (A330)

B. Emergency Equipment System

3. Underwater Locator Beacon (ULB)

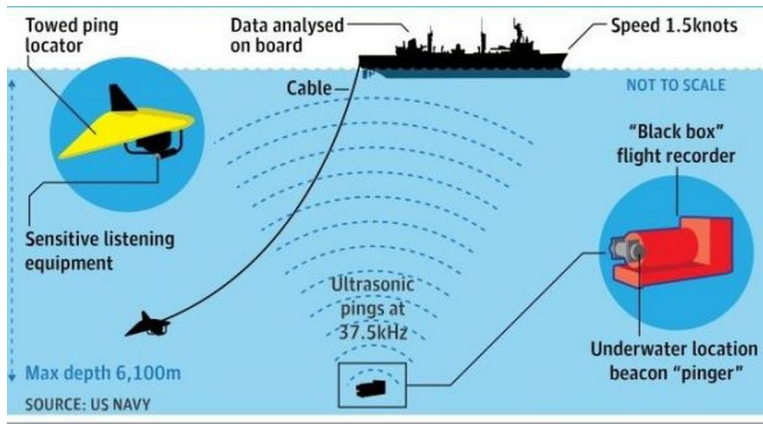
1) Theory of Operation

Beacon is a battery operated underwater acoustic pulse generator that is activated when the water switch is immersed in either fresh or salt water.

Beacon radiates acoustic energy into the surrounding water at 37.5 kHz.

The pulses generated are approximately 10 milliseconds in duration, and occur about once per second in water.

The beacon will operate for **a minimum of 90 days** after being immersed in the water.



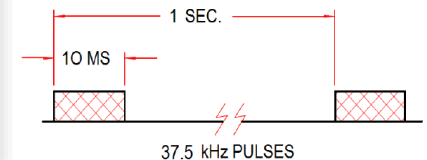
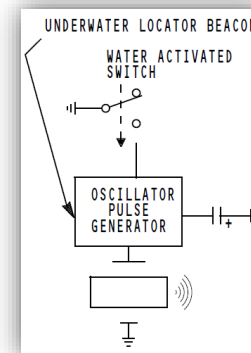
특징

- 발신주파수 : 37.5 khz 음파
- 작동기간 : 30일->90일
- Battery 수명 : 6년

작동조건

- 감지범위 * 1,800~3,600m
- 최대 20,000피트(6,096m)의 깊이 견딤

Note. 해상 상태, 근처 보트, 해양 동물, 가스 또는 석유 라인, 주변 소음 수준에 영향을 미치는 기타 요인은 비콘을 감지할 수 있는 범위에 영향을 미칩니다.



❖ 23 COMMUNICATION SYSTEM (A330)

B. Emergency Equipment System

4. Emergency Locator Transmitter (ELT)

1) General

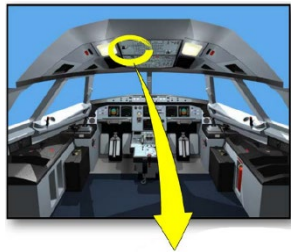
The Emergency Locator Transmitter (ELT) function is to transmit distress signals. It can be used in automatic mode installed on board the aircraft or manually when you use the ELT in portable mode. The ELT system transmits on 3 frequencies homing-signals for civil and military, and to satellite system.

2) ELT Operation

An automatic operation is triggered when the ELT is connected to its system in the A/C, and the G-switch detects an impact sufficient to start the transmission.

A manual operation occurs when the ELT is connected to its system in the A/C and the G-switch is not triggered.

- set to the ON position for manual operation of the ELT.



[Automatic Fixed ELT]

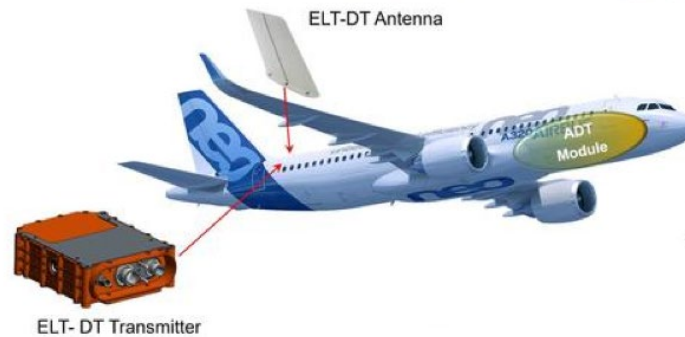
조난 주파수

- 121.5, 243 & 406 MHz
- (406MHz에 국적 및 항공기 정보 등이 저장되어 있으며 매 50초 마다 440ms 동안 위성으로 발신됨)

작동조건

- 자동작동 (가속 Sensor가 5g 감지 시)
- 수동작동(Switch 작동)
- Antenna가 수면 위에 위치해야 발신된 조난 주파수 감지됨

Airbus solution - Equipment overview on A320 / A330 / A350 and A220 families



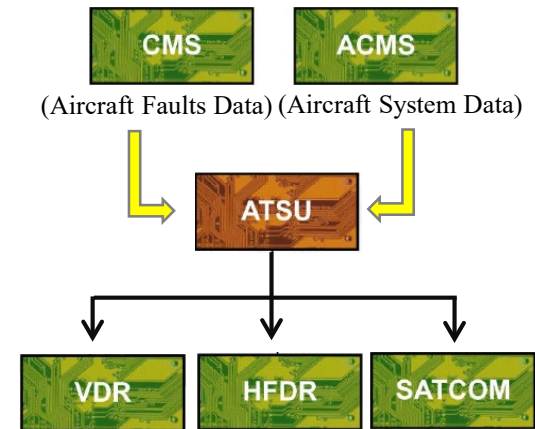
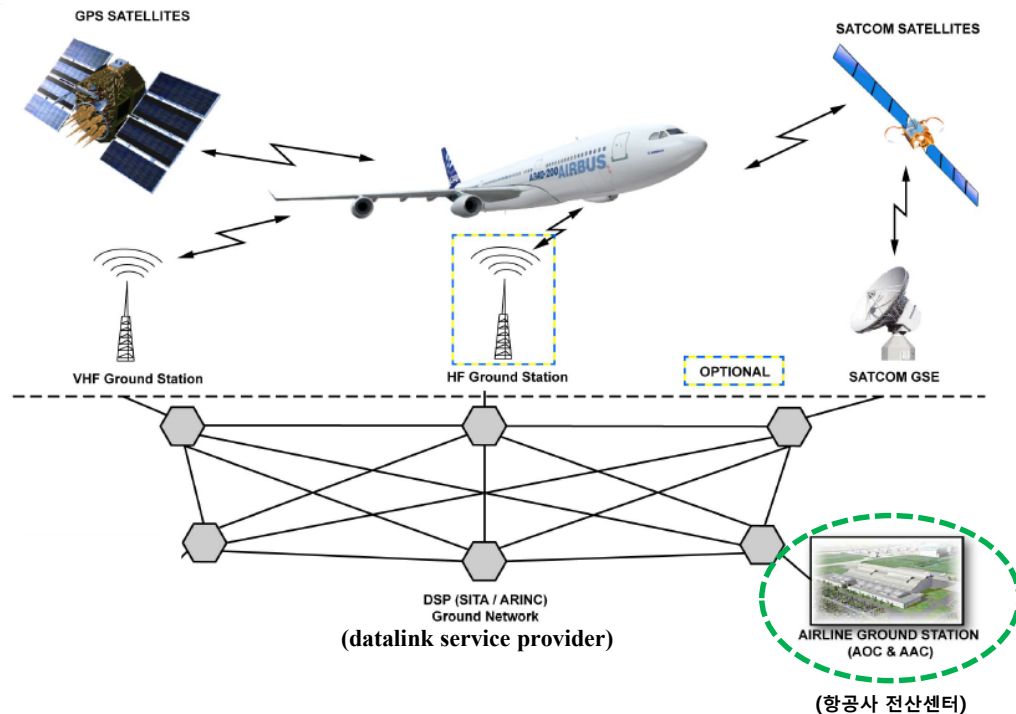
[기내 보관 Survival (Portable) ELT]

❖ 46 INFORMATION SYSTEM (A330)

❖ DATALINK

1) ACARS (정비/항공사 운영 Data 전송)

- **ACARS (Aircraft Communication Addressing and Reporting System)** is a digital data-link system that is used for communication between the airplane and the ground station..
- The ARINC communication addressing and reporting system (ACARS) does these functions:
 - Collects data from airplane systems (Aircraft Faults Data & Aircraft System Data),
 - Sends tuning data to a VHF(or HF) transceiver,
 - Sends data to the SATCOM satellite data unit (SDU),
 - Air Traffic Service Unit (ATSU) for direct exchange of data between the aircraft and a ground network.

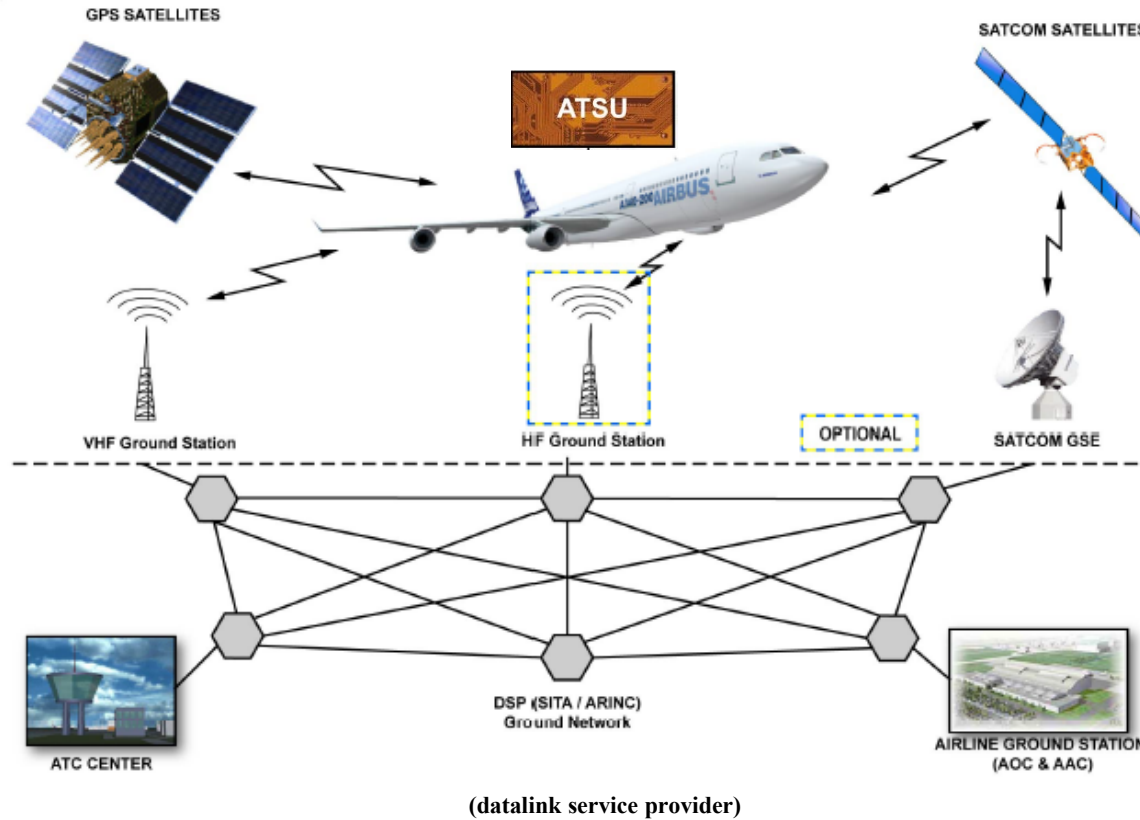


- Central maintenance computer (CMC)
- Aircraft Condition Monitoring System (ACMS)
- Air Traffic Service Unit (ATSU)

❖ 46 INFORMATION SYSTEM (A330)

The Information Systems send data-link transfers between the aircraft and the ground. This chapter is divided into:

- the DATALINK,



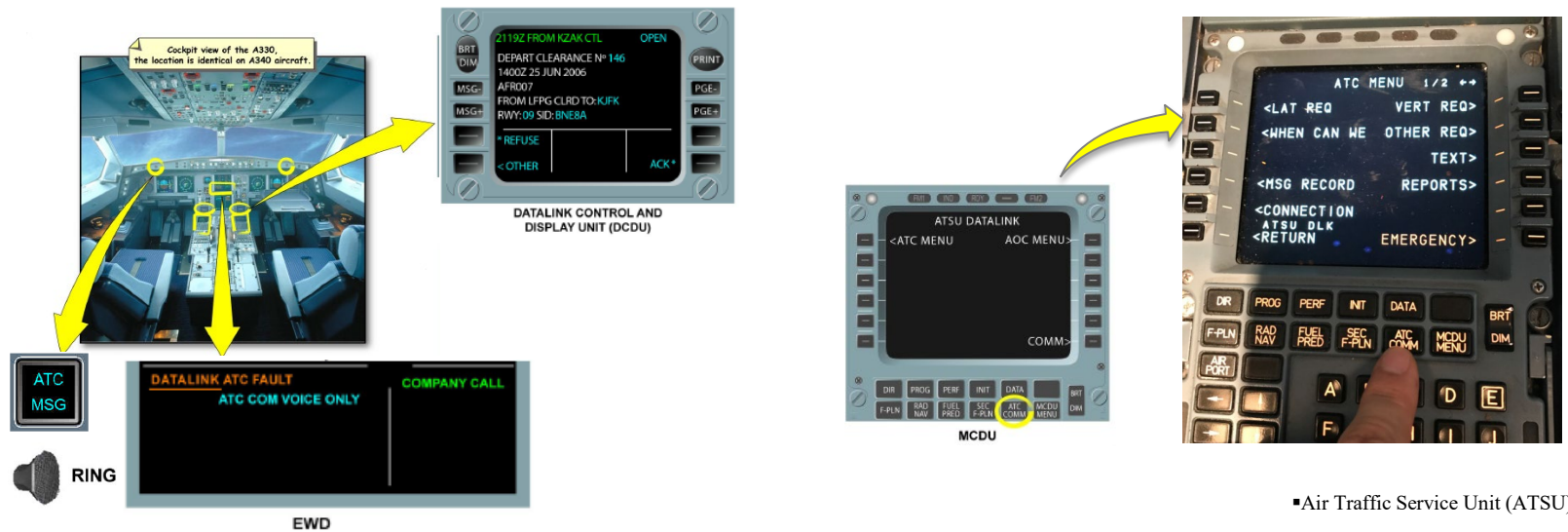
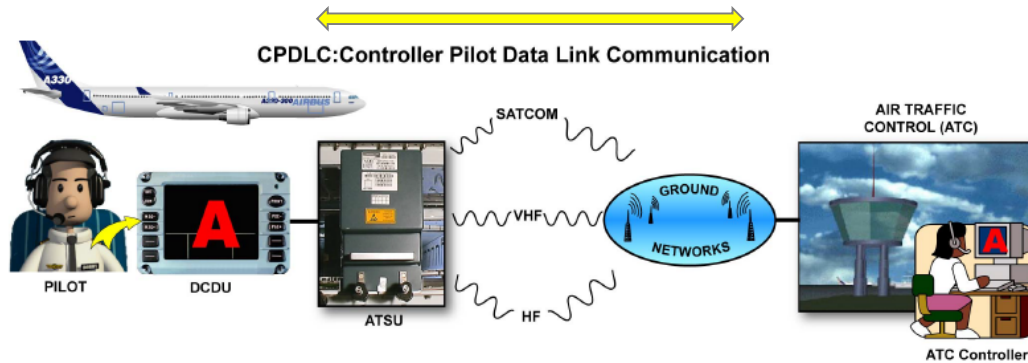
▪ Air Traffic Service Unit (ATSU)

❖ 46 INFORMATION SYSTEM (A330)

❖ DATALINK

2) CPDLC (관제사와 DATA 통신)

- The CPDLC application enables an exchange of specific ATC messages between the flight crew and air traffic controller.
- Voice 관제에서 Data Link를 기반으로 한 관제 가능



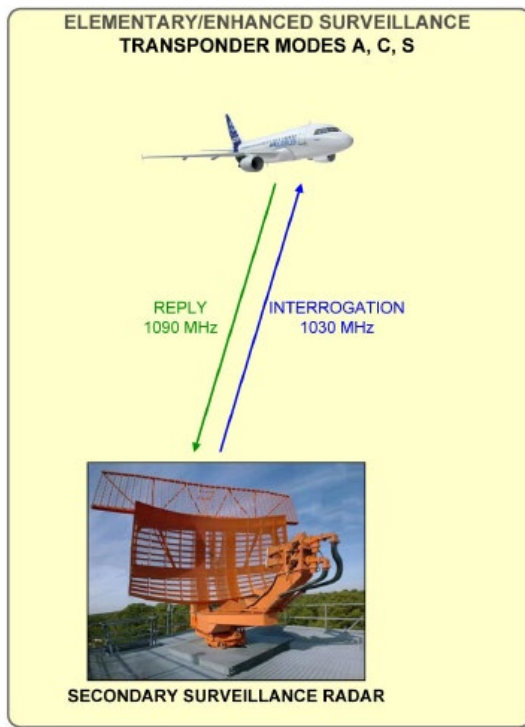
■ Air Traffic Service Unit (ATSU)

❖ 46 INFORMATION SYSTEM (A330)

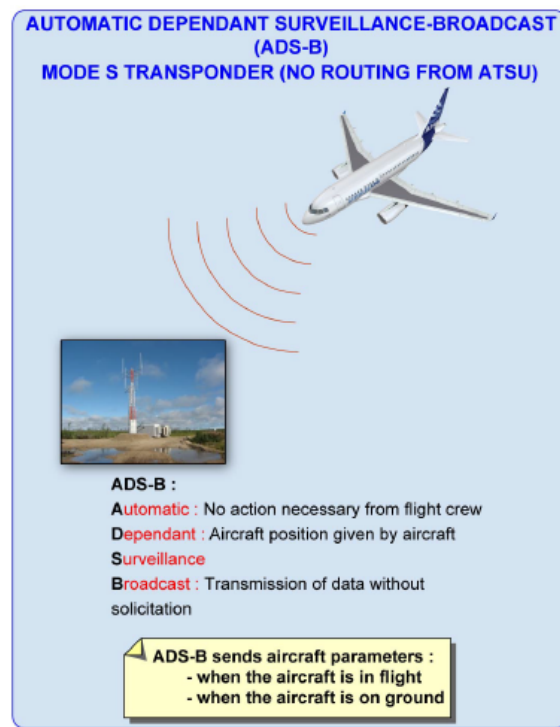
❖ DATALINK

3) NAVIGATION AND SURVEILLANCE FOR AIR TRAFFIC MANAGEMENT (항공 관제 감시 시스템)

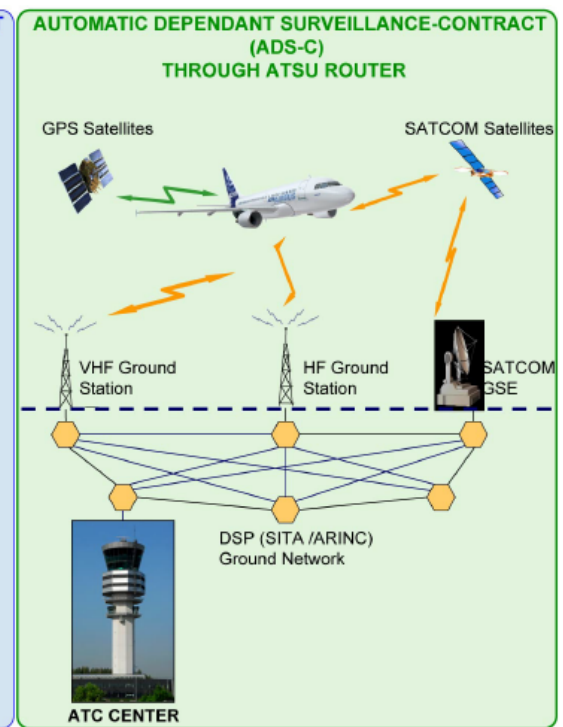
- **The Automatic Dependent Surveillance (ADS) system** sends aircraft position and aircraft navigation data to ATC centers and other aircraft.
- There are two different ADS applications:
 - ADS-Contract (ADS-C) automatically sends aircraft surveillance data to connected ATC centers via ATC Datalink in remote or oceanic areas.
 - ADS-Broadcast (ADS-B) automatically broadcasts the aircraft position and navigation data to other users (ATC centers or other aircraft) equipped with a Mode S transponder.



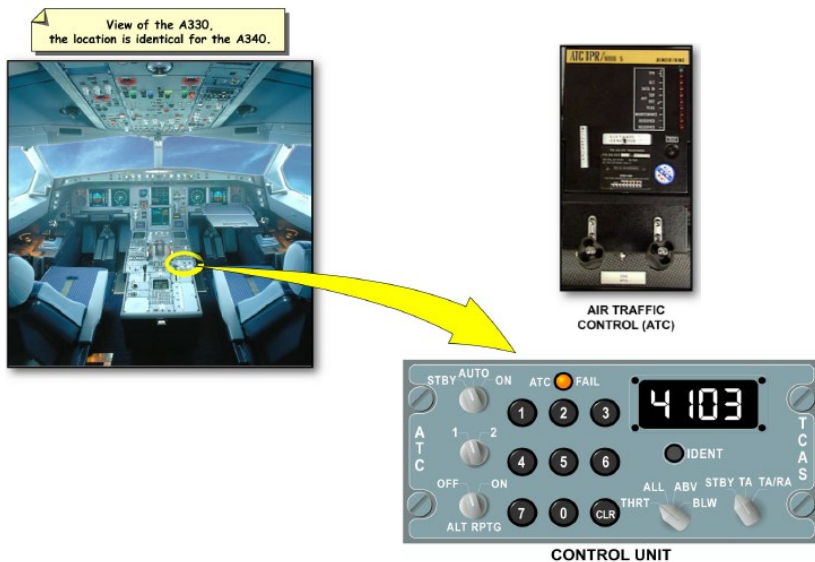
ELEMENTARY SURVEILLANCE (ELS)



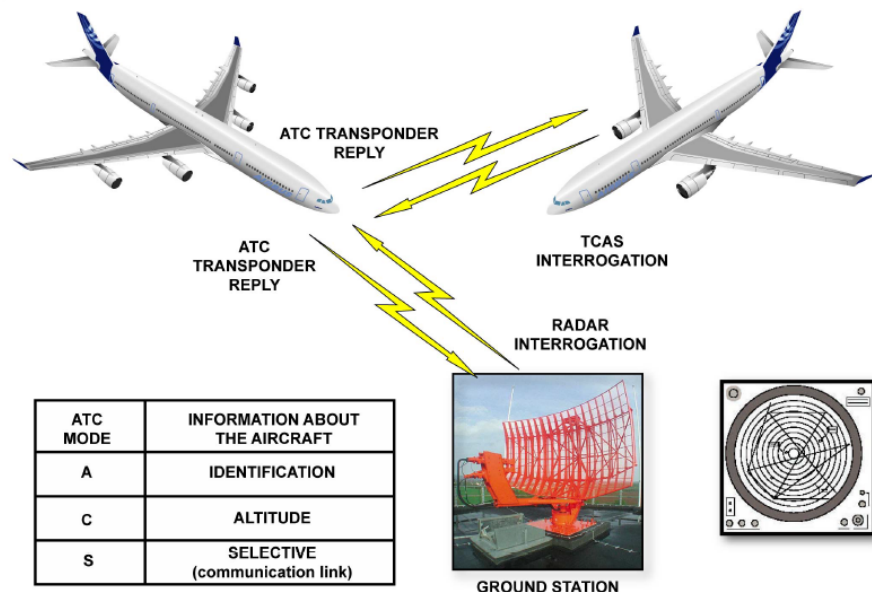
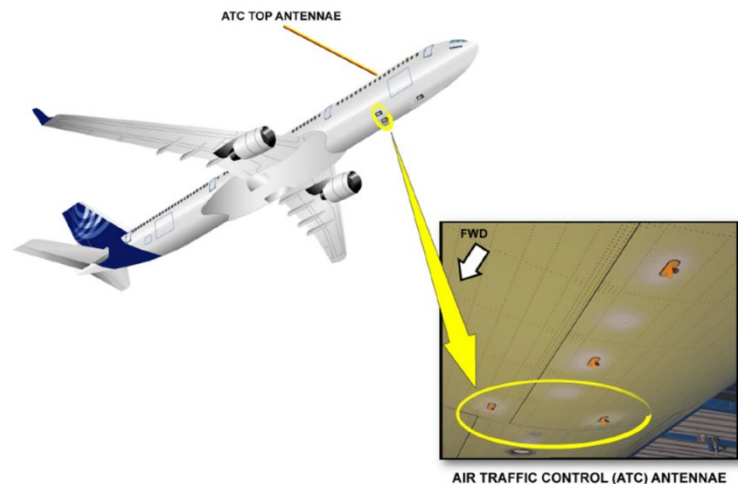
AUTOMATIC DEPENDANT SURVEILLANCE (ADS)



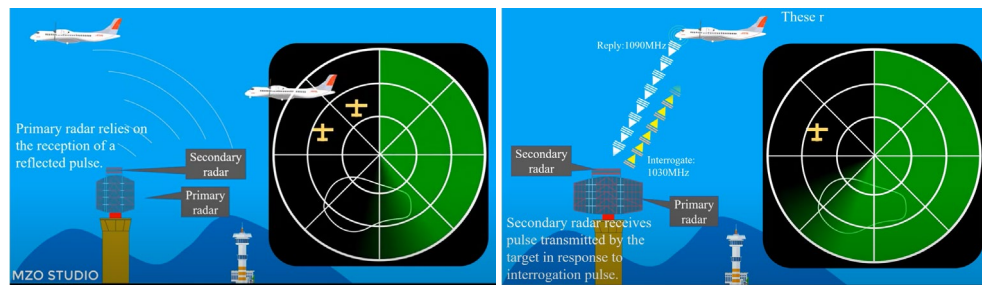
❖ 46 INFORMATION SYSTEM (A330)



ATC/TCAS - ATC/TCAS CONTROL



ATC/TCAS - ATC PRINCIPLE



❖ 46 INFORMATION SYSTEM (A330)

AUTOMATIC DEPENDANT SURVEILLANCE-BROADCAST (ADS-B) MODE S TRANSPONDER (NO ROUTING FROM ATSU)



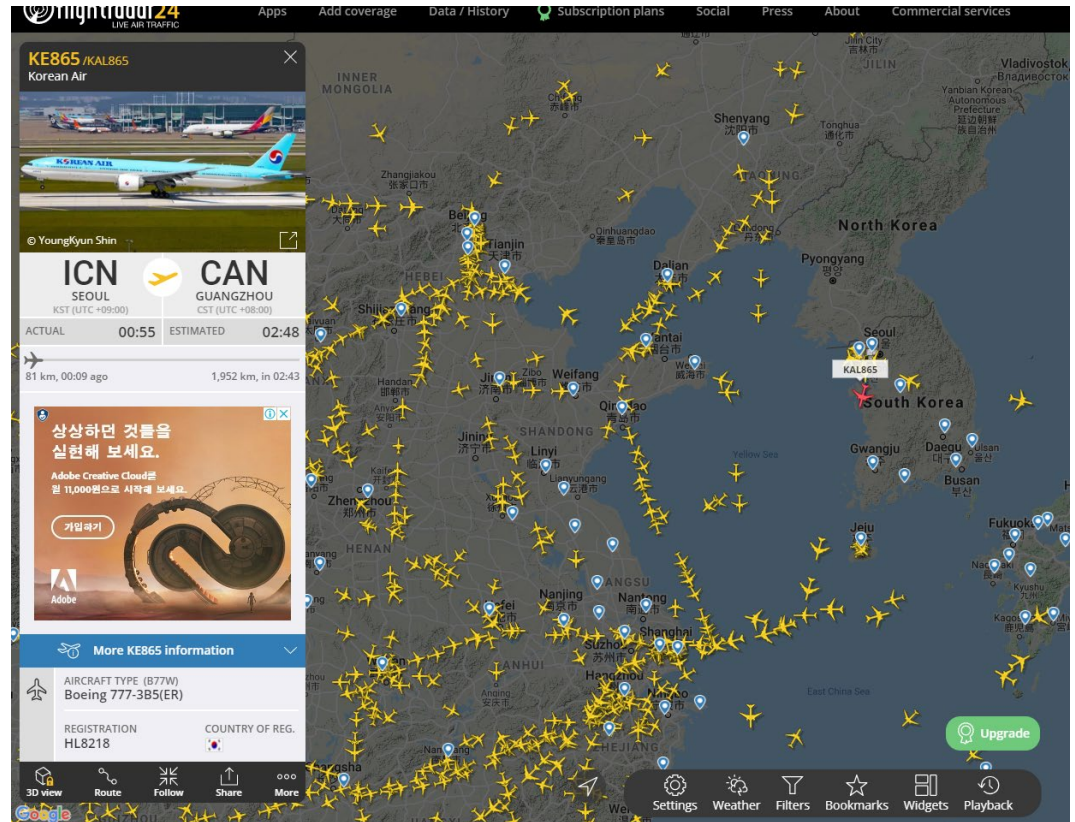
ADS-B :

- Automatic** : No action necessary from flight crew
- Dependant** : Aircraft position given by aircraft
- Surveillance**
- Broadcast** : Transmission of data without solicitation

ADS-B sends aircraft parameters :
- when the aircraft is in flight
- when the aircraft is on ground



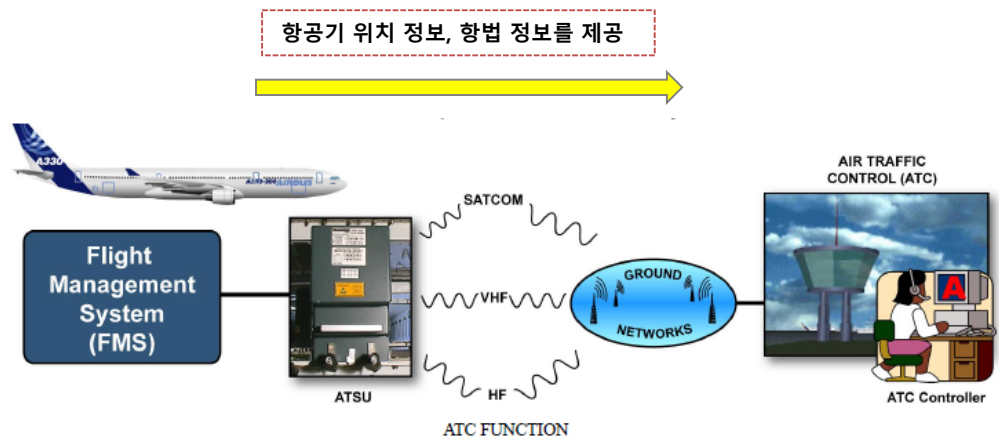
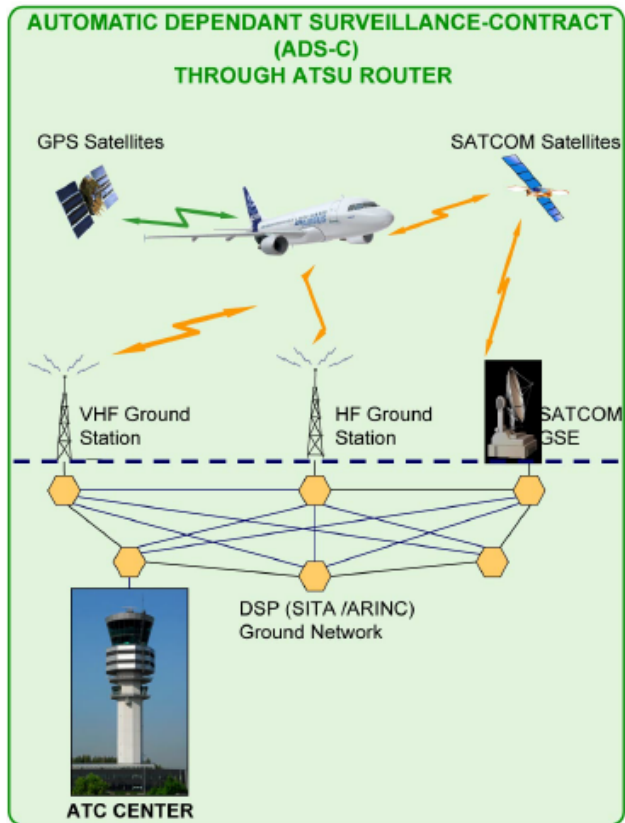
AIR TRAFFIC
CONTROL (ATC)



<https://www.flightradar24.com>

ADS-Broadcast (ADS-B) – Data 관제

❖ 46 INFORMATION SYSTEM (A330)



▪ Air Traffic Service Unit (ATSU)

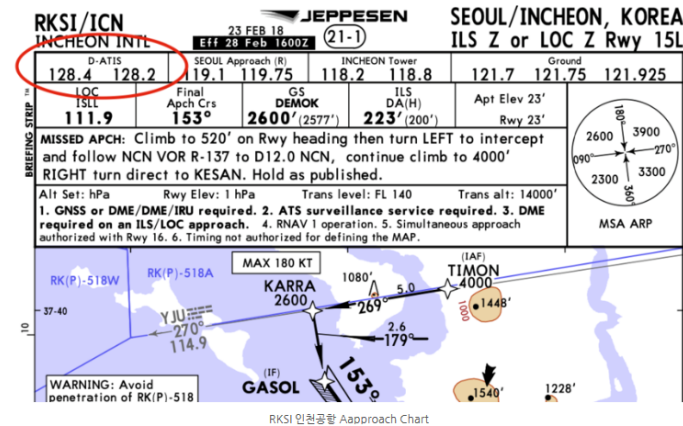
❖ ADS-Contract (ADS-C) – Data 관제

❖ 46 INFORMATION SYSTEM (A330)

❖ DATALINK

4) D-ATIS (공항에 대한 정보수신)

- The Digital Automatic Terminal Information Service (D-ATIS) enables the flight crew to request and obtain information about airports.
- VHF channel의 Voice Information 에서 Data Link 기반으로 한 ATIS 적용.



[PRINTER 출력 가능]



[화면상으로 보는 Arrival / Departure 메시지]

❖ 34 NAVIGATION SYSTEM (A330)

- The navigation system is composed of **several sub-systems**, which are:
 - the Air Data/Inertial Reference System (ADIRS),
 - the Integrated Standby Instrument System (ISIS),
 - **the Multi-Mode Receiver (MMR),**
 - the Radio Altimeter (RA),
 - the Enhanced Ground Proximity Warning System (EGPWS),
 - **the Weather Radar (WXR) with Predictive WindShear (PWS) capability,**
 - the Air Traffic Control (ATC),
 - the Traffic Alert and Collision Avoidance System (TCAS),
 - **the VOR and MKR beacon,**
 - **the Distance Measuring Equipment (DME),**
 - the Automatic Direction Finder (ADF) if installed,
 - the Digital Distance and Radio Magnetic Indicator (DDRMI),
 - the maintenance/test facilities.

A. ILS

The aircraft has two Multi Mode Receivers (MMRs). The MMR is a single navigation receiver that includes two internal receivers:

- **an ILS receiver &** a GPS receiver.

The primary function of the MMR is to receive and process ILS and GPS signals..

B. WEATHER RADAR

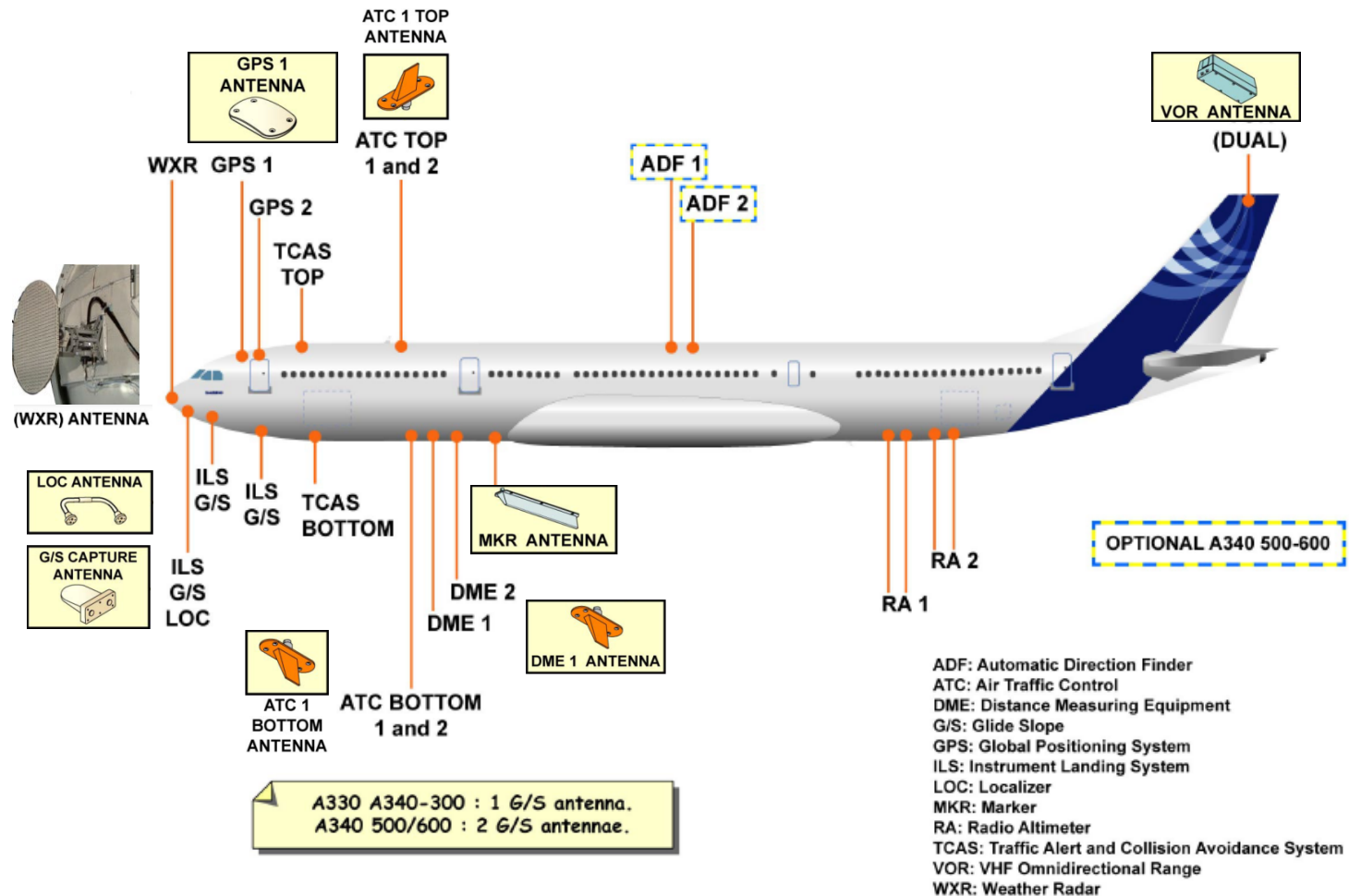
The airborne WXR and PWS is used for the detection and display of severe weather areas, gives a ground mapping and also detects windshear events.

C. VOR / MARKER / DME

The Very high frequency Omni directional and radio Range (VOR), Distance Measuring Equipment (DME), and MARKER systems are radio navigation aids, which compute navigation data (bearing, slant distance) from ground stations.

❖ 34 NAVIGATION SYSTEM (A330)

- This diagram shows the location of the various radio-navigation antenna.

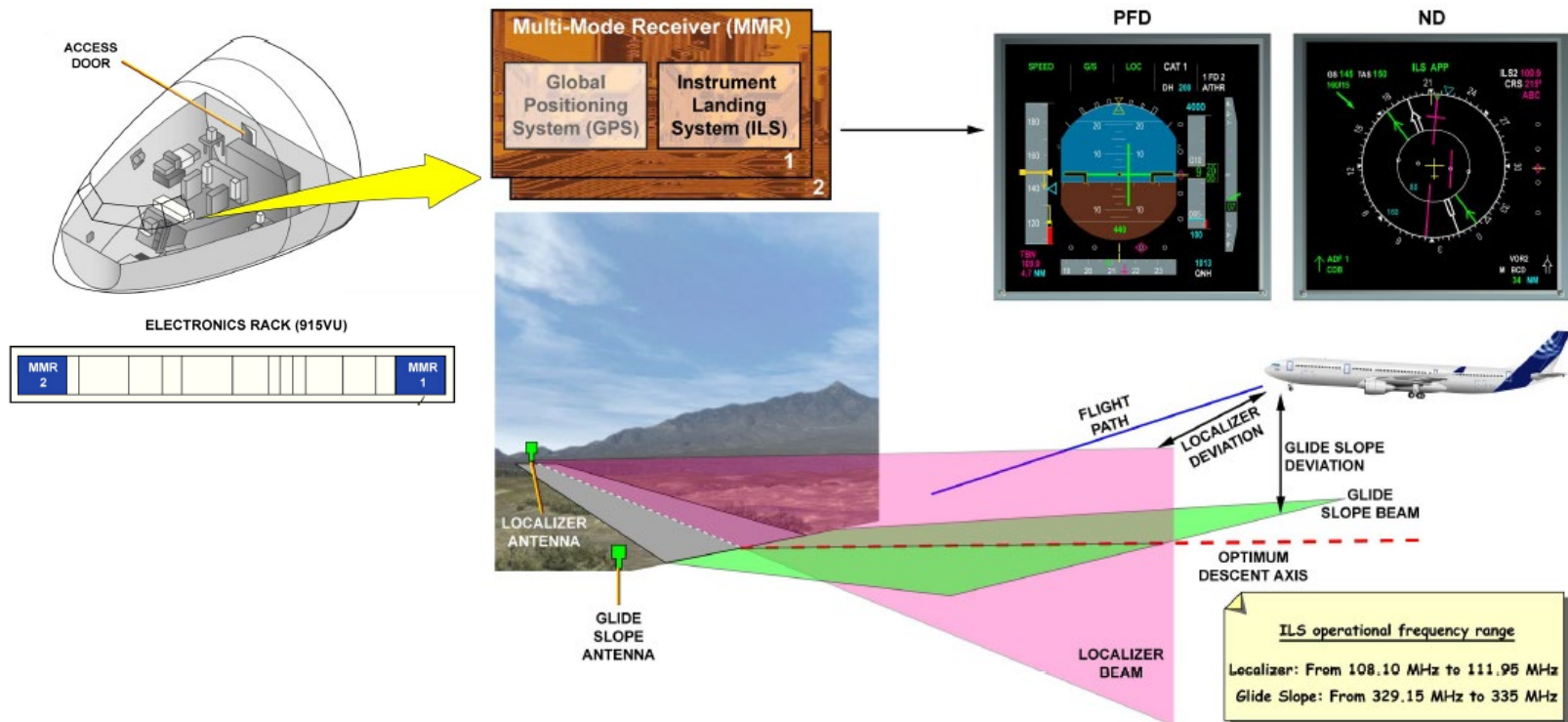


❖ 34 NAVIGATION SYSTEM (A330)

A. ILS

1) ILS PRINCIPLE

The ILS has ground stations sending information to the crew and aircraft systems to guide the A/C along a predetermined flight path for the landing phase. The airborne ILS receiver measures and displays angular deviations with respect to this predetermined flight path and decodes the morse audio signal, which identifies the ground station. A descent axis is determined by the intersection of a localizer beam (LOC) and a Glide Slope beam (G/S) generated by the ground stations.



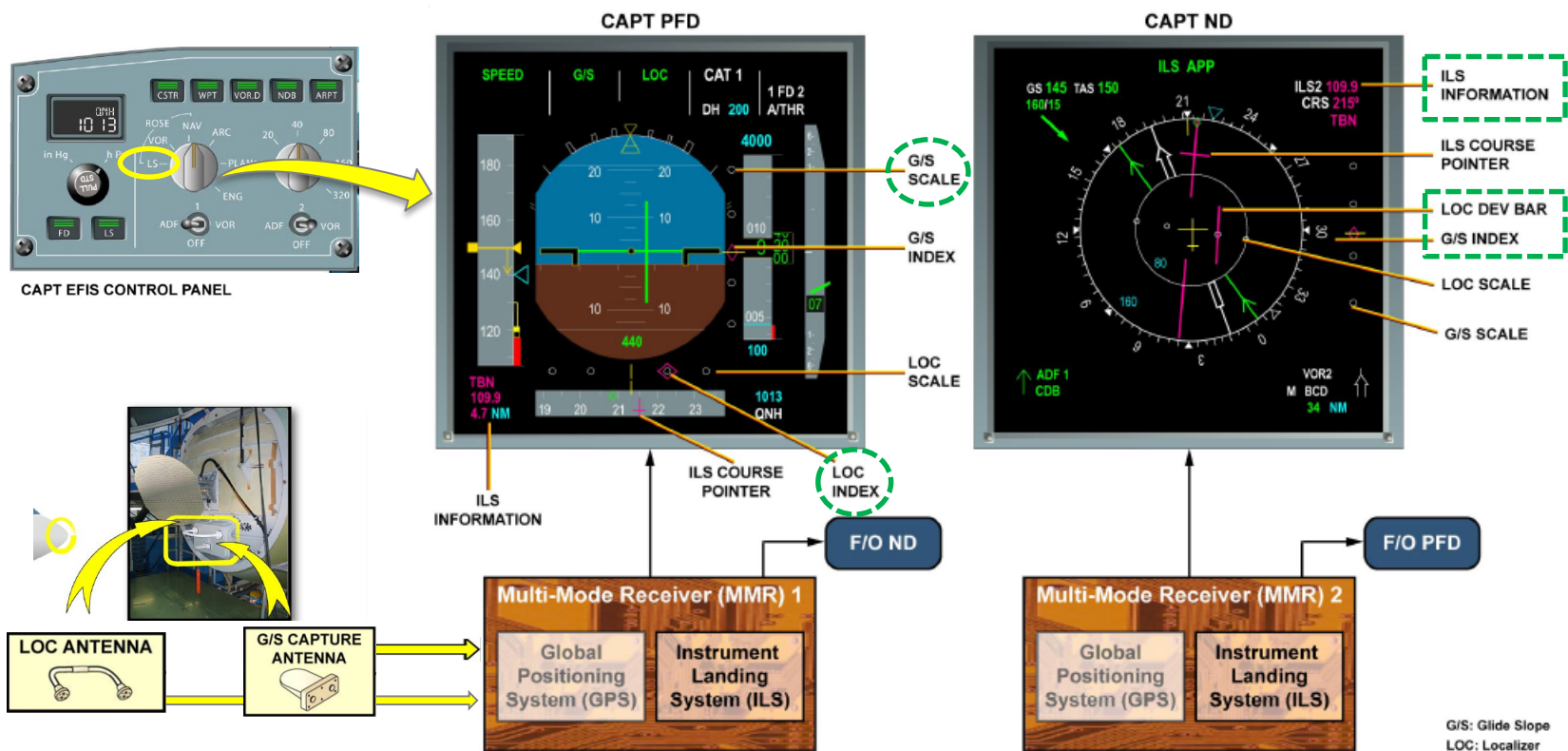
- Instrument Landing System (ILS)
- Primary Flight Display
- Navigation Display

❖ 34 NAVIGATION SYSTEM (A330)

A. ILS

2) ILS INDICATING

The ILS data is sent to the displays through the Display Management Computers (DMCs). The ILS 1 data is sent to the CAPT PFD and F/O ND, while the ILS 2 data is sent to the F/O PFD and CAPT ND.



❖ 34 NAVIGATION SYSTEM (A330)

B. WEATHER RADAR

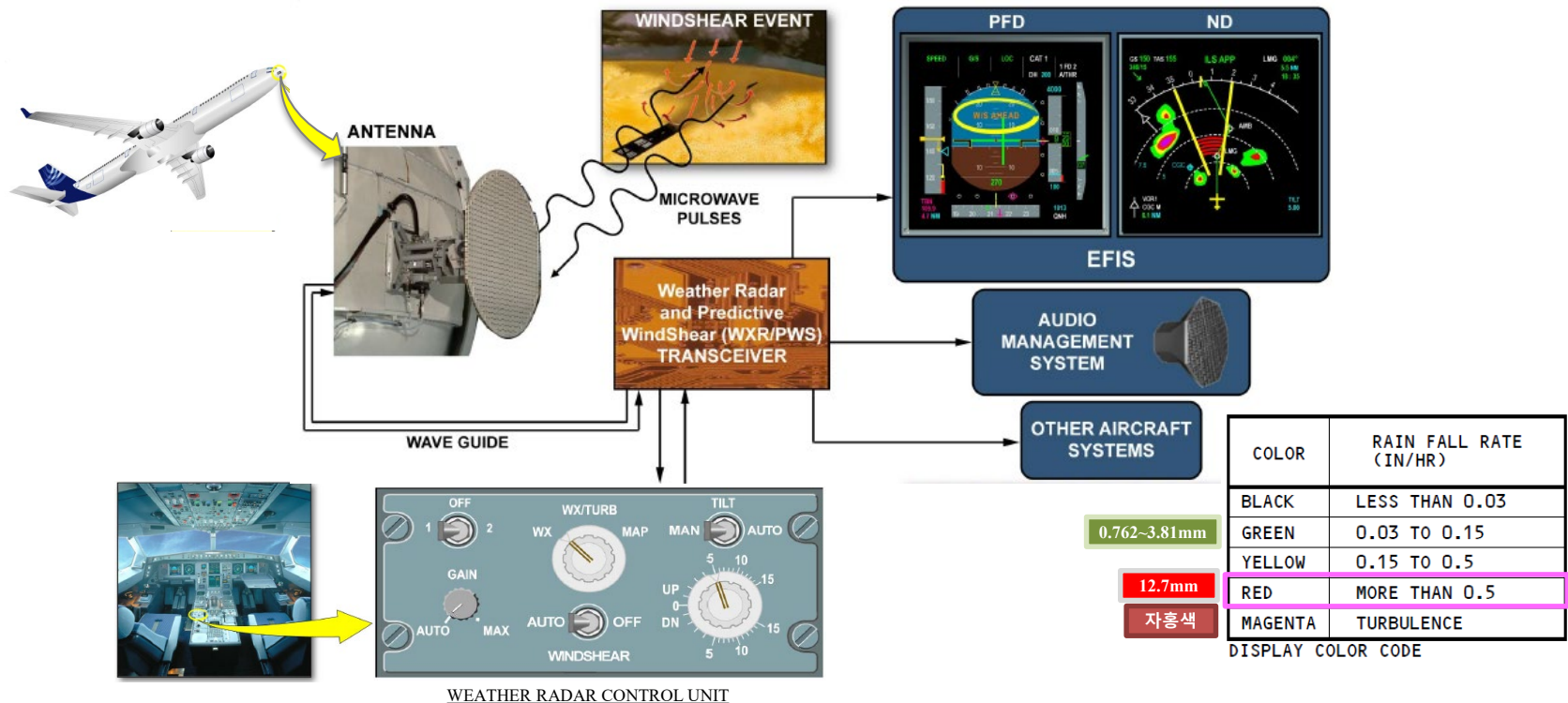
1) WEATHER RADAR COMPONENT LOCATION

The weather Radar control panel is installed on the aft part of the pedestal.

The antenna is in the nose cone. Access to the antenna is done by the radome. The transceivers are on the forward part of the avionics compartment (left hand side).

2) WEATHER RADAR CONTROL AND INDICATING

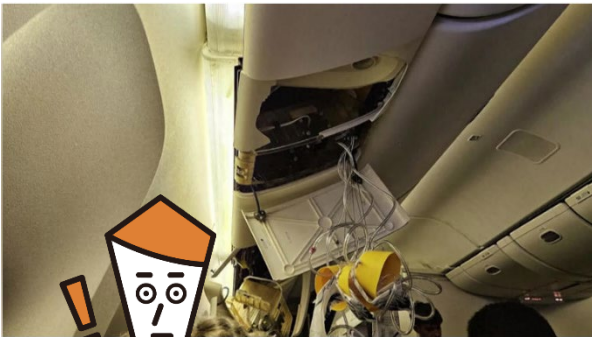
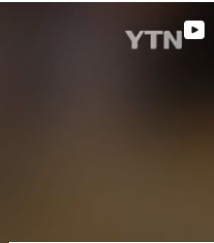
The WXR/PWS transceiver generates the microwave pulses. A wave-guide is used as a Radio Frequency (RF) connection between the transceiver and the antenna. The antenna is used for transmitting and receiving the RF signals. The return signal is processed and formatted by the transceiver for display on the ND. When a windshear event is detected, the information is displayed to the crew on both EFIS displays. An aural warning is also generated. The WXR/PWS transceiver also interfaces with other aircraft systems. A weather radar control unit enables the crew to control the mode of operation of the WXR/PWS. It also enables them to adjust several parameters (e.g. antenna tilt or gain control).



❖ 기상 이변

"난기류 비상착륙"

승객들에 따르면 갑자기 비행기가 기울고 급격하게 하강하면서 안전벨트를 착용하지 않은 사람들이 수하물 칸 등 기내 곳곳에 부딪히며 아수라장이 벌어졌습니다.



항로 추적 웹사이트인 '플라이트 레이더24'를 보면 고도 37,000피트에서 순항하던 여객기가 3분 만에 31,000피트까지 6천 피트, 1,800미터 이상 급강하했습니다.

❖ 34 NAVIGATION SYSTEM (A330)

C. VOR / MARKER / DME

1) GENERAL

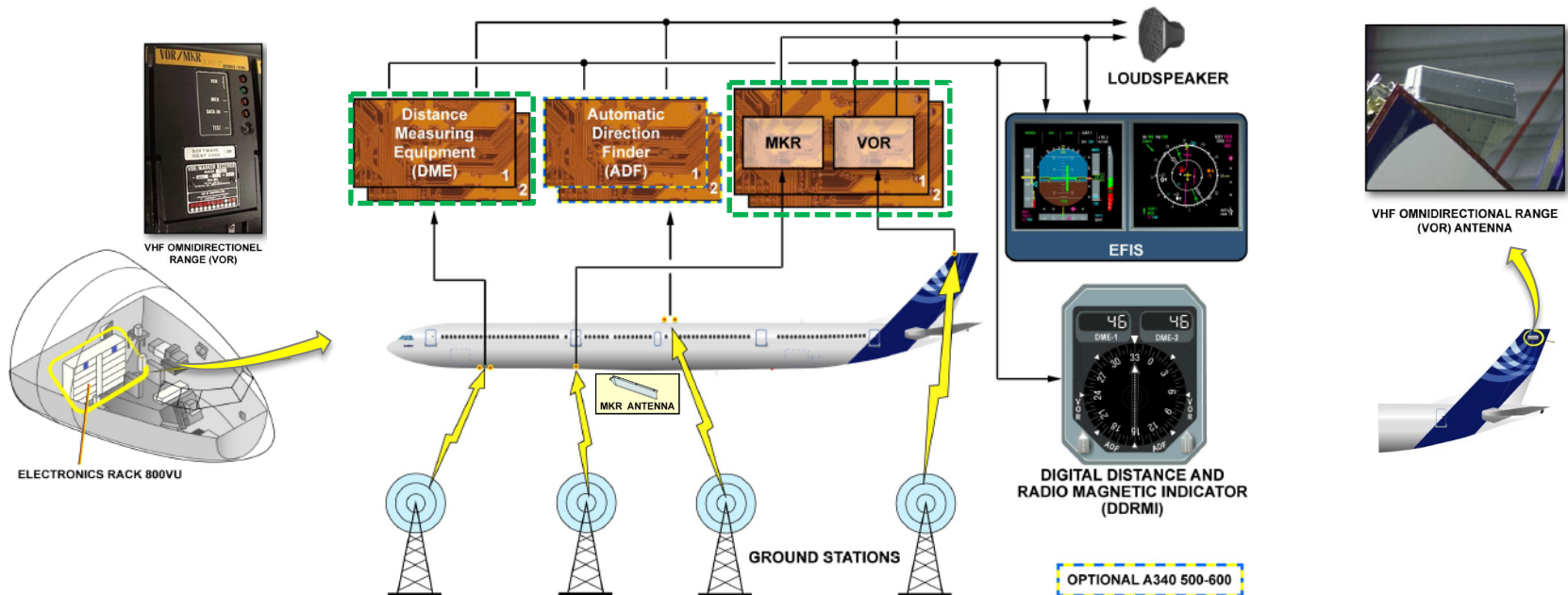
The VOR system is a navigation aid that receives, decodes and processes signals received from the omnidirectional ground station, for the bearing information. The MKR system is a radio navigation aid, which gives indicators related to the distance between the aircraft and the runway threshold. When the aircraft over flies one of the MKR transmitters, the system gives aural and visual indications to the flight crew.

NOTE: The MKR and VOR features are included in the same computer.

The MKR function is only active in VOR/MKR 1. The DME system is a radio navigation aid that computes the slant distance between the aircraft and a DME ground station. ADF computes the bearing of ADF GND station. The ADF system is a radio navigation aid that computes the bearing of an ADF ground station in respect of the aircraft heading. Data are displayed to the crew.

2) VOR COMPONENT LOCATION

The VOR antenna is installed on the top of the vertical fin. To have access to this antenna, the vertical stabilizer tip (324AT) must be removed. The VOR computers are installed on the 800VU electronic rack.



❖ 34 NAVIGATION SYSTEM (A330)

C. VOR / MARKER / DME

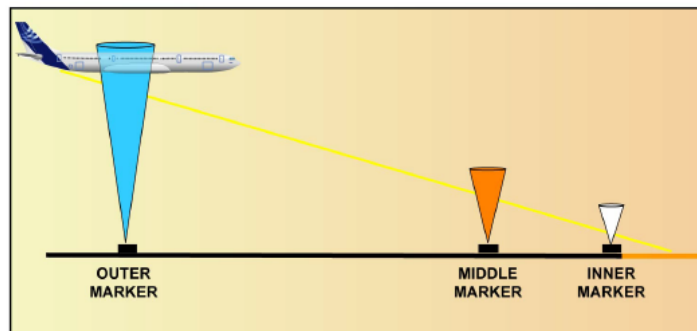
3) CONTROL AND INDICATING

VOR/DME/ADF data are displayed on the ND according to the EFIS control panels selection. MKR information is displayed on CAPT and F/O PFDs.



CAPT EFIS CONTROL PANEL

ND



The marker function is only active in VOR/MKR 1.

▪ Primary Flight Display (PFD)



▪ Navigation Display (ND)

❖ 34 NAVIGATION SYSTEM (A330)

D. FREQUENCY TUNING ILS

1) ILS / VOR / MARKER / DME

① AUTO TUNING

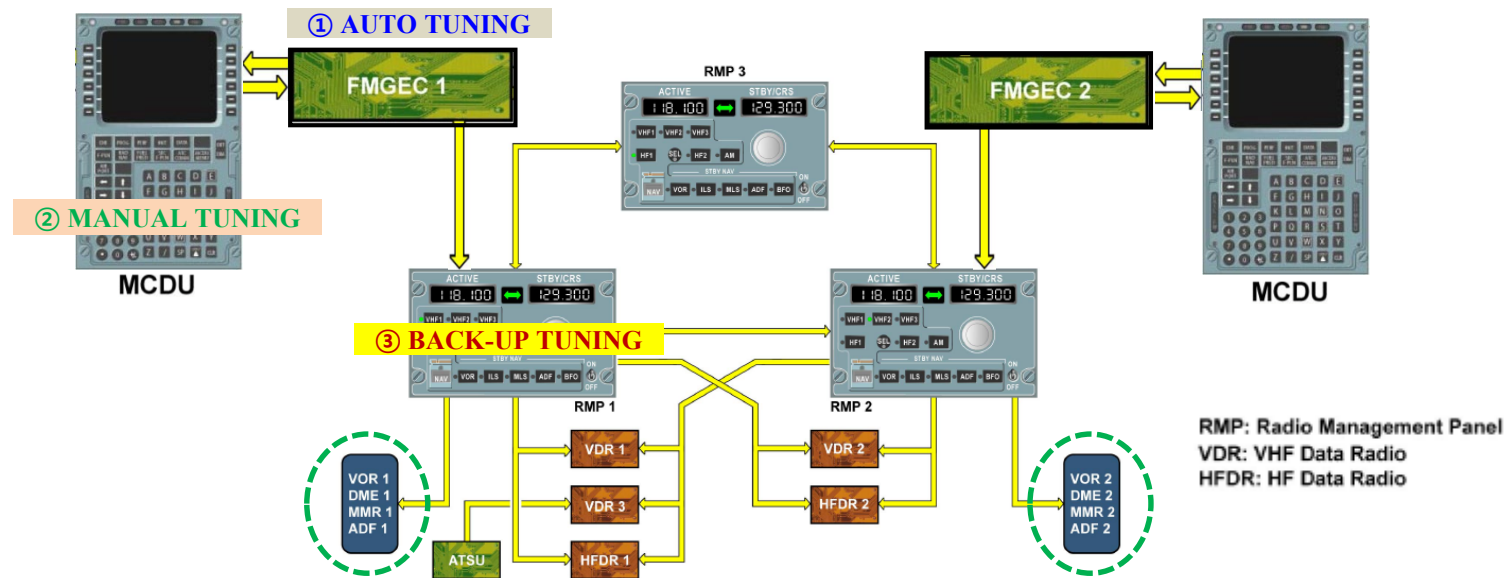
In normal operation each Flight Management Guidance and Envelope Computer (**FMGEC**) automatically tunes its onside ILS receiver through its onside Radio Management Panel (RMP).

② MANUAL TUNING

Each MCDU allows the onside ILS receiver to be manually tuned through the onside FMGEC and the opposite ILS receiver to be tuned through the opposite FMGEC.

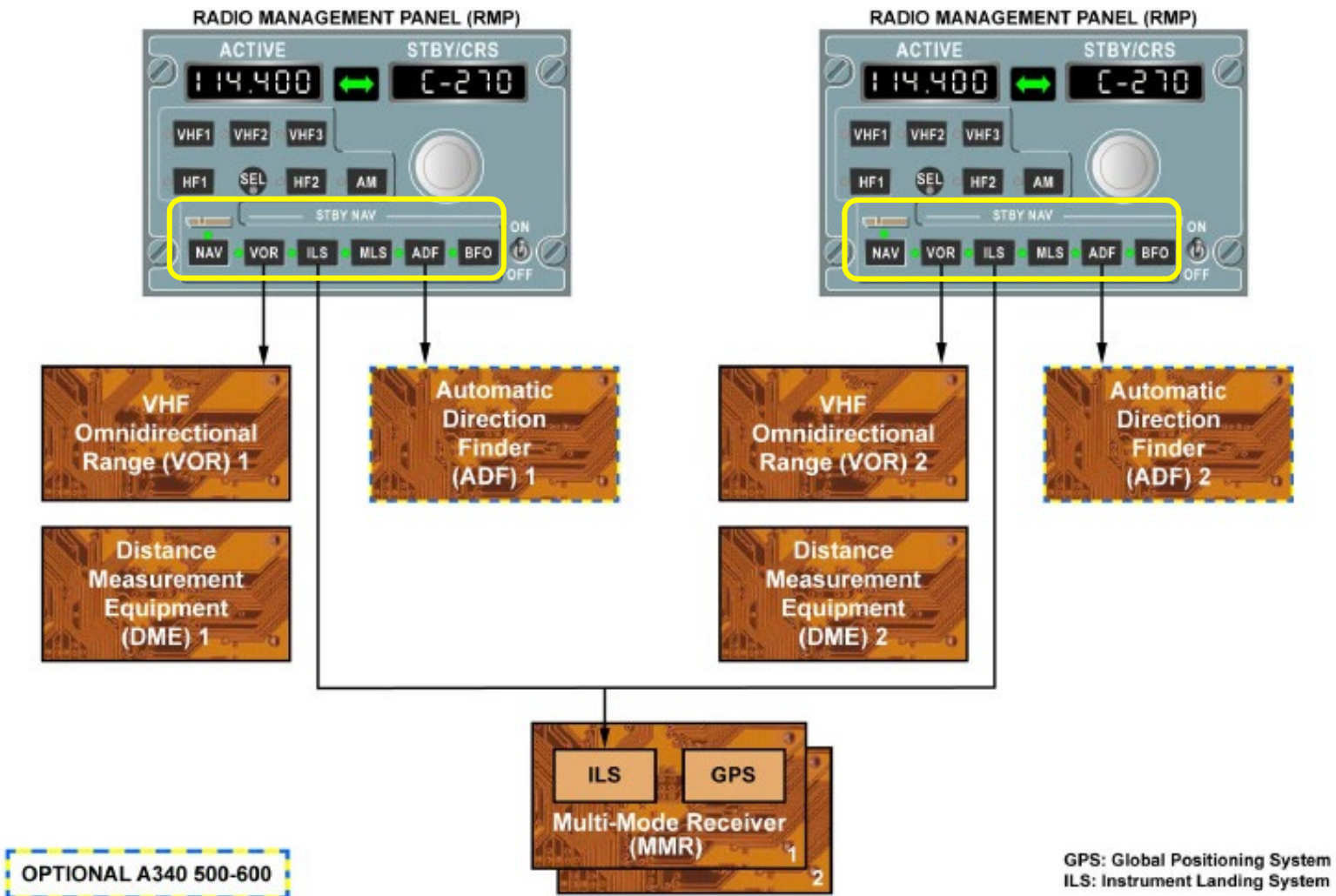
③ BACK-UP TUNING

If both FMGECs fail, each RMP allows its onside receiver to be directly tuned.



- Instrument Landing System (ILS)
- Maintenance Control Display Unit(MCDU)

❖ 34 NAVIGATION SYSTEM (A330)



RADIO NAVIGATION CONTROL - BACKUP TUNING

Q & A

